

5ME42A(ME) - Manufacturing Technology

For 5th Semester B.E. Mechanical Engineering

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Metal Forming

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Broad classification of Engineering Manufacturing Processes.

1. Shaping or forming

■ Manufacturing a solid product of definite size and shape from a given material taken in three possible states:

- in liquid or semi-liquid state – e.g., casting, injection moulding etc.
 - in solid state – e.g., forging rolling, extrusion, drawing etc.
 - in powder form – e.g., powder metallurgical process.
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2. Joining process

- Welding, brazing, soldering etc.

3. Removal or Cutting process

- Machining (Traditional or Non-traditional), Grinding etc.

4. Regenerative manufacturing Process

Production of solid products in layer by layer from raw materials in different form:

- ❑ liquid – e.g., stereo lithography
- ❑ powder – e.g., selective sintering
- ❑ sheet – e.g., LOM (laminated object manufacturing)
- ❑ wire – e.g., FDM. (Fused Deposition Modeling)

It is the latest one which is generally accomplished very rapidly and quite accurately using CAD and CAM for Rapid Prototyping and Tooling.

METAL FORMING

1. Overview of Metal Forming
2. Material Behavior in Metal Forming
3. Temperature in Metal Forming
4. Friction and Lubrication in Metal Forming

Metal Forming

Large group of manufacturing processes in which plastic deformation is used to change the shape of metal workpieces

- The tool, usually called a *die*, applies stresses that exceed the yield strength of the metal
 - The metal takes a shape determined by the geometry of the die

Metal Forming

Plastic deformation increases hardness and strength while ductility is decreased known as ***Work Hardening***

Stresses in Metal Forming

- Stresses to plastically deform the metal are usually compressive
 - Examples: rolling, forging, extrusion
- However, some forming processes
 - Stretch the metal (tensile stresses)
 - Others bend the metal (tensile and compressive)
 - Still others apply shear stresses

Material Properties in Metal Forming

- Desirable material properties:
 - Low yield strength
 - High ductility
- These properties are affected by temperature:
 - Ductility increases and yield strength decreases when work temperature is raised
- Other factors:
 - Strain rate and friction

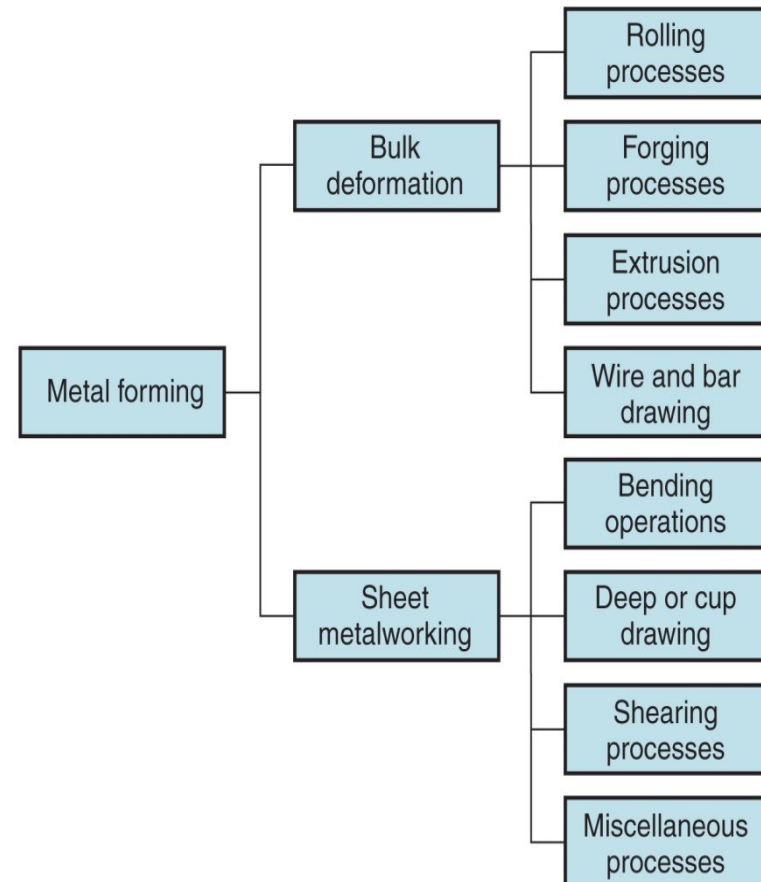
Metal Forming Processes

1. Bulk deformation

- Rolling processes
- Forging processes
- Extrusion processes
- Wire and bar drawing

2. Sheet metalworking

- Bending operations
- Deep or cup drawing
- Shearing processes

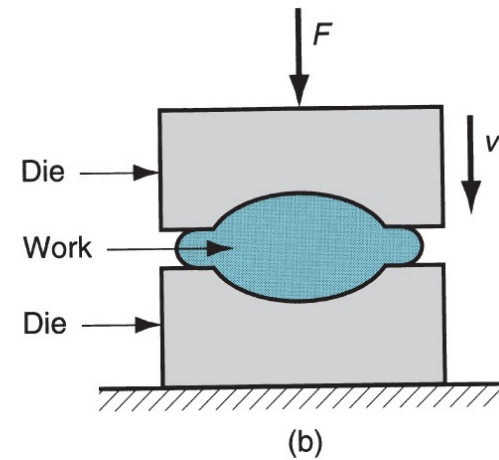
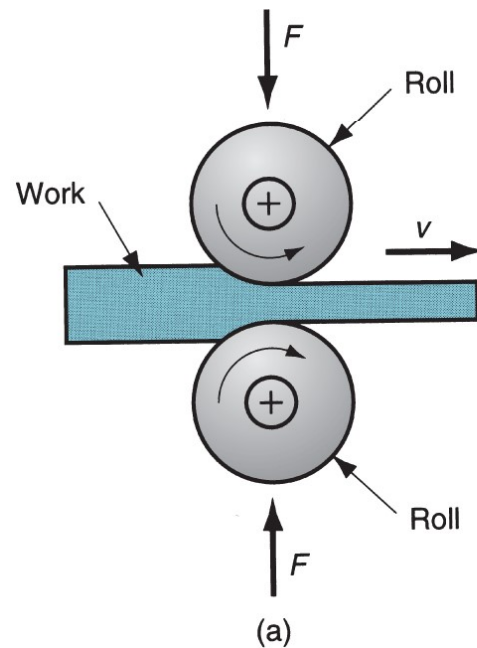


Bulk Deformation Processes

- Characterized by significant deformations and massive shape changes
- "Bulk" refers to work parts with relatively low surface area-to-volume ratios
- Starting work shapes are usually simple geometries
 - Examples:
 - Cylindrical billets
 - Rectangular bars

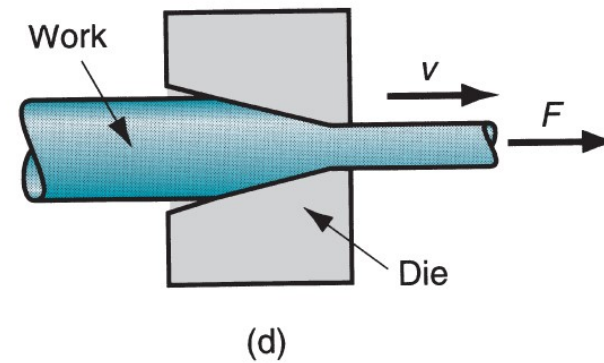
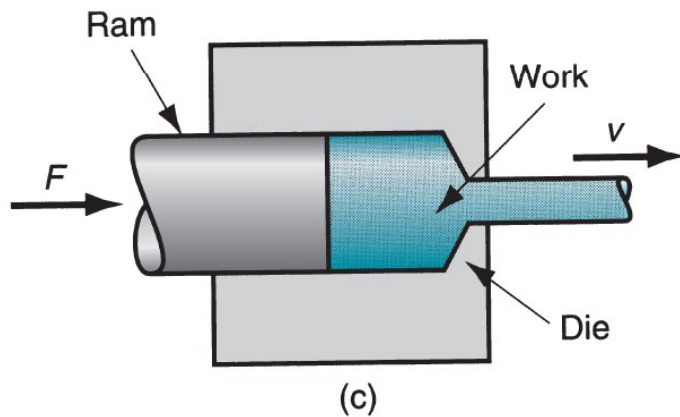
Bulk Deformation Processes

- (a) Rolling and (b) forging



Bulk Deformation Processes

- (c) Extrusion and (d) wire and bar drawing

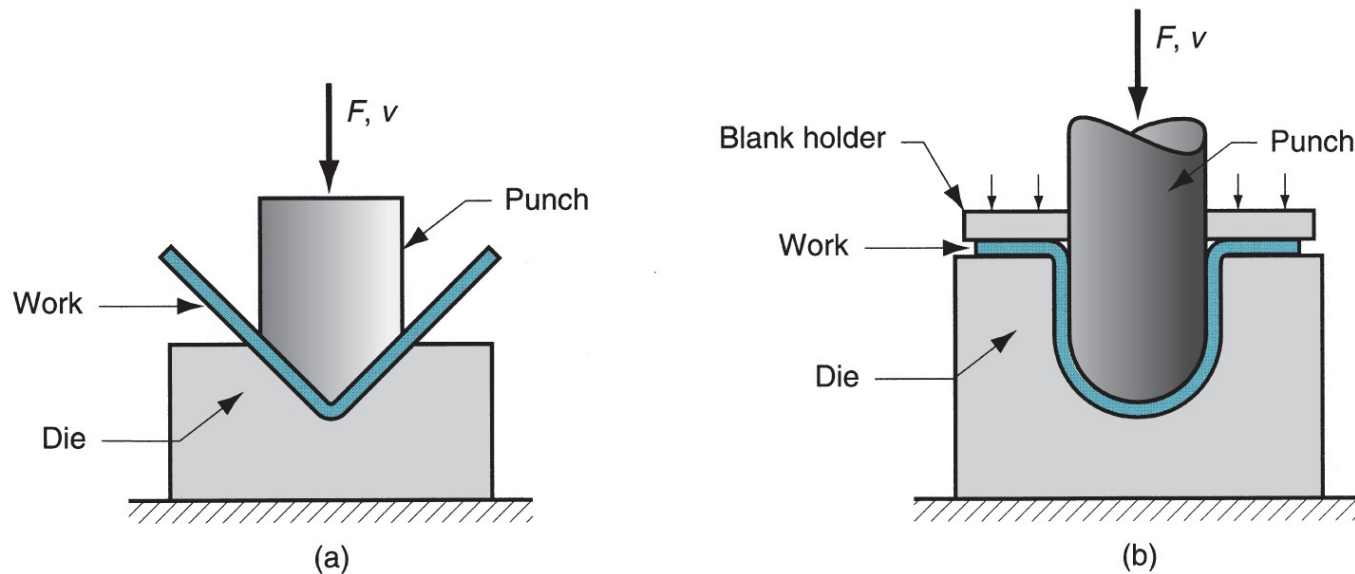


Sheet Metalworking

- Forming and related operations performed on metal sheets, strips, and coils
- High surface area-to-volume ratio of starting metal, which distinguishes these from bulk deformation
- Often called *pressworking* because these operations are performed on presses
 - Parts are called *stampings*
 - Usual tooling: *punch* and *die*

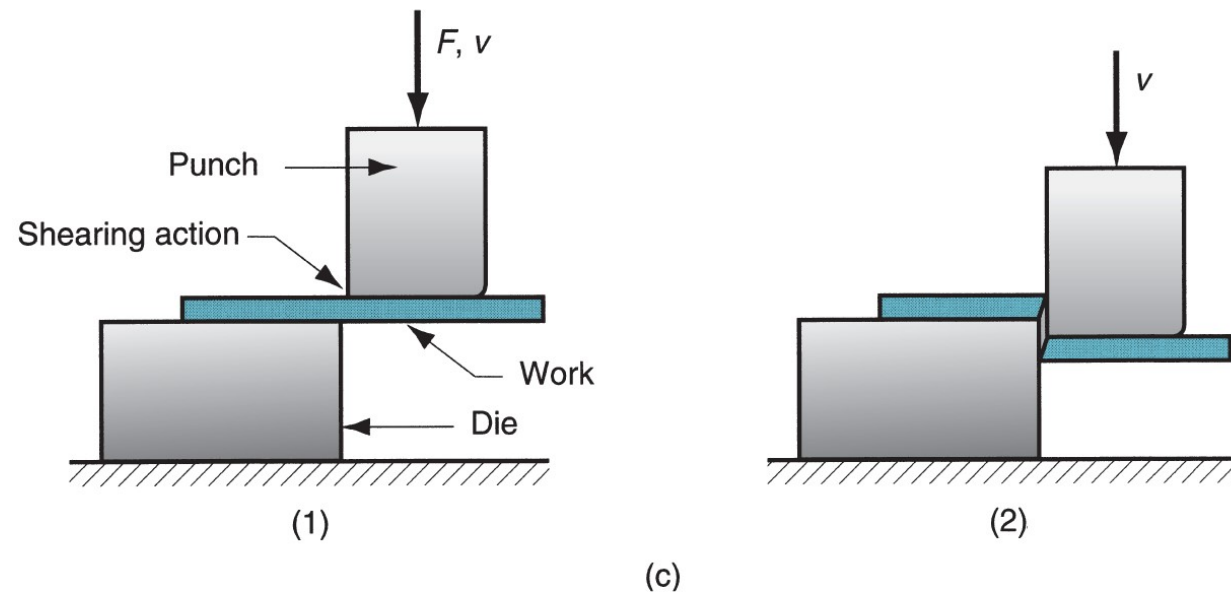
Sheet Metalworking

- (a) Bending and (b) deep drawing



Sheet Metalworking

- (c) Shearing: (1) punch first contacting sheet and (2) after cutting



Material Behavior in Metal Forming

- Plastic region of stress-strain curve is primary interest because material is plastically deformed
- In plastic region, metal's behavior is expressed by the flow curve:

$$\sigma = K\varepsilon^n$$

where K = strength coefficient; and n = strain hardening exponent

Flow Stress

- For most metals at room temperature, strength increases when deformed due to strain hardening
- *Flow stress* = instantaneous value of stress required to continue deforming the material

$$Y_f = K\varepsilon^n$$

where Y_f = flow stress, that is, the yield strength as a function of strain

Average Flow Stress

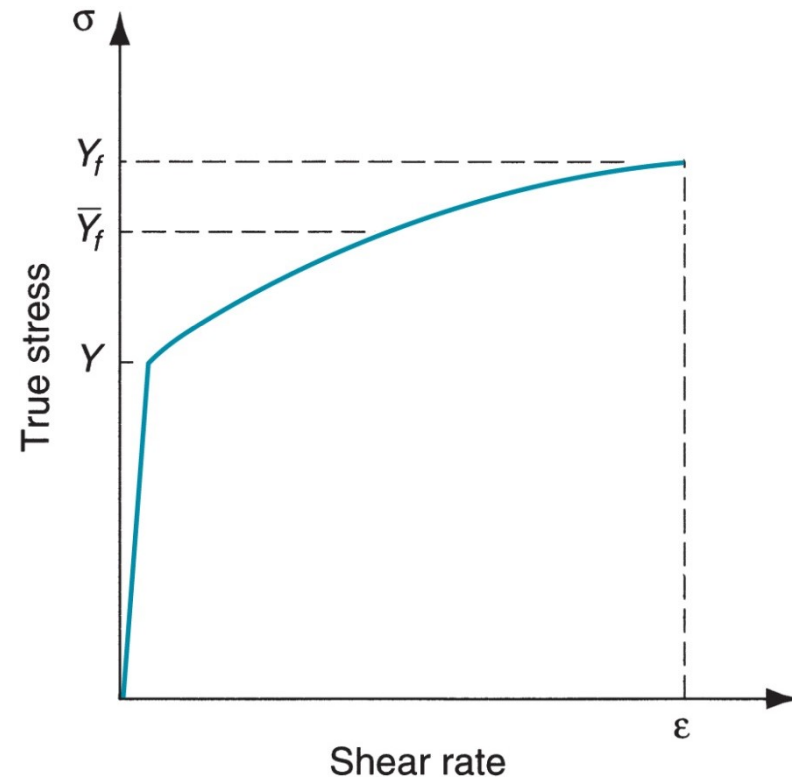
- Determined by integrating the flow curve equation between zero and the final strain value defining the range of interest

$$\bar{Y}_f = \frac{K\varepsilon^n}{1+n}$$

where $\bar{Y}_f$ = average flow stress; and ε = maximum strain during deformation process

Stress-Strain Relationship

- Average flow stress $\bar{Y}_f$ in relation to
 - Flow stress Y_f
 - Yield strength Y



Temperature in Metal Forming

- For any metal, K and n in the flow curve depend on temperature
 - Both strength (K) and strain hardening (n) are reduced at higher temperatures
 - In addition, ductility is increased at higher temperatures

$$\sigma = K\varepsilon^n$$

Temperature in Metal Forming

- Any deformation operation can be accomplished with lower forces and power at elevated temperature
- Three temperature ranges in metal forming:
 - Cold working
 - Warm working
 - Hot working

Cold Working

- Performed at room temperature or slightly above
- Many cold forming processes are important mass production operations
- Minimum or no machining usually required
 - These operations are *near net shape* or *net shape* processes

Advantages of Cold Forming

- Better accuracy, closer tolerances
- Better surface finish
- Strain hardening increases strength and hardness
- Grain flow during deformation can cause desirable directional properties in product
- No heating of work required

Disadvantages of Cold Forming

- Higher forces and power required for deformation
- Surfaces of starting work must be free of scale and dirt
- Ductility and strain hardening limit the amount of forming that can be done
 - In some cases, metal must be annealed before further deformation can be accomplished
 - In other cases, metal is simply not ductile enough to be cold worked

Warm Working

- Performed at temperatures above room temperature but below recrystallization temperature
- Dividing line between cold working and warm working often expressed in terms of melting point:
 - $0.3T_m$, where T_m = melting point (absolute temperature) for metal

Advantages and Disadvantages of Warm Working

- Advantages
 - Lower forces and power than in cold working
 - More intricate work geometries possible
 - Need for annealing may be reduced or eliminated
- Disadvantage
 - Workpiece must be heated

Hot Working

- Deformation at temperatures above the *recrystallization temperature*
 - Recrystallization temperature = about one-half of melting point on absolute scale
 - In practice, hot working usually performed somewhat above $0.5T_m$
 - Metal continues to soften as temperature increases above $0.5T_m$, enhancing advantage of hot working above this level

Why Hot Working?

Capability for substantial plastic deformation - far more than is possible with cold working or warm working

- Why?
 - Strength coefficient (K) is substantially less than at room temperature
 - Strain hardening exponent (n) is zero (theoretically)
 - Ductility is significantly increased

Advantages of Hot Working

- Workpart shape can be significantly altered
- Lower forces and power required
- Metals that usually fracture in cold working can be hot formed
- Strength properties of product are generally isotropic
- No strengthening of part occurs from work hardening
 - Advantageous in cases when part is to be subsequently processed by cold forming

Disadvantages of Hot Working

- Lower dimensional accuracy
- Higher total energy required, which is the sum of
 - The thermal energy needed to heat the workpiece
 - Energy to deform the metal
- Work surface oxidation (scale)
 - Thus, poorer surface finish
- Shorter tool life
 - Dies and rolls in bulk deformation

Friction in Metal Forming

- In most metal forming processes, friction is undesirable:
 - Metal flow is reduced
 - Forces and power are increased
 - Tools wear faster
- Friction and tool wear are more severe in hot working

Lubrication in Metal Forming

- Metalworking lubricants are applied to tool-work interface in many forming operations to reduce harmful effects of friction
- Benefits:
 - Reduced sticking, forces, power, tool wear
 - Better surface finish
 - Removes heat from the tooling

Considerations in Choosing a Lubricant

- Type of forming process (rolling, forging, sheet metal drawing, etc.)
- Hot working or cold working
- Work material
- Chemical reactivity with tool and work metals
- Ease of application
- Cost